

- 30** A radioactive source is placed 1 cm from a Geiger-Muller tube, and various absorbers are placed between them, one at a time. For each absorber, a one-minute count is taken of the total number of decays, and this is repeated several times. The table shows the results of the experiment.

absorber	average number of decays detected in one minute
none	1043
0.1 mm paper	1040
1 mm aluminium	497
1 cm lead	6

The average background count per minute is 5.

What nuclear radiation do the results suggest the source was emitting?

- A** alpha and beta only
- B** beta only
- C** beta and gamma only
- D** alpha, beta and gamma

Answer: B

Type of decay	alpha	beta	gamma
Penetration power	3 cm of air A sheet of paper	3 m of air A sheet of aluminum of a few mm thickness	Lead, concrete or steel of >10 cm thickness

Because most particles can pass through paper, they are not alpha particles.

Some of them are absorbed by the aluminium, and stopped by 4 cm lead; they must be beta particles.

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